A ZARISKI LOCALLY QUASI-SEPARATED ALGEBRAIC SPACE CONTAINS A DENSE OPEN SUBSPACE WHICH IS A SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to a single affine chart with the properties of the two projections, the construction of the invariant open over which the source projection is finite by generic finiteness and descent, and the passage to the quotient of that open reducing the statement to the finite etale case.

Editorial note. The label proposition-locally-quasi-separated-open-dense-scheme names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-properties.tex –label proposition-locally-quasi-separated-open-dense-scheme –start 1863 –end 1906 –local-id 28048 returned RESULT=OK (L28048 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Local nature of the statement in the space. Same chapter, the lemma that the property of containing a dense open subscheme can be checked Zariski locally. Supplies the reduction to a single piece of a Zariski cover.
- Characters of the charts given by the local quasi-separatedness. Same chapter, the lemma characterizing Zariski local quasi-separatedness through quasi-compact surjective etale morphisms from affine schemes. Supplies the affine chart U -> X, which is separated because the cover is separated.
- Generic finiteness, descent of finiteness and invariance of the locus of finiteness. Morphisms, the lemma producing an open neighbourhood of a generic point over which a quasi-finite morphism becomes finite, Descent, the lemma that finiteness descends along fpqc morphisms, and More on Groupoids, the lemma that the largest open locus where the source map is finite is invariant. Supply the invariant open W containing all generic points of irreducible components of U, over which the relation is finite.
- Restriction of the relation to an invariant open and the finite etale case. Groupoids, the description of the restriction of a groupoid to an invariant open, and Same chapter, the lemma treating the case of a finite etale cover.

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Supply the open subspace X' = W/R_W and the final reduction to the finite etale case.

Source reference. Original lemma 02LA: sources/stacks-chow/tags.txt, source lines 3401–3401. Prerequisite pointer invoked at proof line(s) 1889; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 03BC: sources/stacks-chow/tags.txt, source lines 4343–4343. Prerequisite pointer invoked at proof line(s) 1897; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BAS: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 1817–1817. Prerequisite pointer invoked at proof line(s) 1905; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03W7: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 1065–1065. Prerequisite pointer invoked at proof line(s) 1874; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03FZ: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 992–992. Prerequisite pointer invoked at proof line(s) 1877; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03JH: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 1782–1782. Prerequisite pointer invoked at proof line(s) 1873; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03JC: sources/stacks-chow/tags.txt, source lines 4634–4634. Prerequisite pointer invoked at proof line(s) 1892; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03WS: sources/stacks-chow/tags.txt, source lines 5123–5123. Prerequisite pointer invoked at proof line(s) 1882; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02NW: sources/stacks-chow/tags.txt, source lines 3495–3495. Prerequisite pointer invoked at proof line(s) 1885; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02NH: sources/stacks-chow/tags.txt, source lines 3480–3480. Prerequisite pointer invoked at proof line(s) 1884; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01TJ: sources/stacks-chow/tags.txt, source lines 2376–2376. Prerequisite pointer invoked at proof line(s) 1883; it names a standard

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02WU: sources/stacks-chow/tags.txt, source lines 3817–3817. Prerequisite pointer invoked at proof line(s) 1901; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let S be a scheme. Let X be an algebraic space over S. If X is Zariski locally quasi-separated (for example if X is quasi-separated), then there exists a dense open subspace X' of X which is a scheme. More precisely, every point $x \in |X|$ of codimension 0 on X is contained in X'.*

Proof. The question is local on X by Lemma 03JH. Thus by Lemma 03W7 we may assume that there exists an affine scheme U and a surjective, quasi-compact, étale morphism $U \to X$. Moreover $U \to X$ is separated (Lemma 03FZ). Set $R = U \times_X U$ and denote $s, t : R \to U$ the projections as usual. Then s, t are surjective, quasi-compact, separated, and étale. Hence s, t are also quasi-finite and have finite fibres (Morphisms, Lemmas 03WS, 01TJ, and 02NH). By Morphisms, Lemma 02NW for every $\eta \in U$ which is the generic point of an irreducible component of U, there exists an open neighbourhood $V \subset U$ of η such that $s^{-1}(V) \to V$ is finite. By Descent, Lemma 02LA being finite is fpqc (and in particular étale) local on the target. Hence we may apply More on Groupoids, Lemma 03JC which says that the largest open $W \subset U$ over which s is finite is R-invariant. By the above W contains every generic point of an irreducible component of U. The restriction R_W of R to W equals $R_W = s^{-1}(W) = t^{-1}(W)$ (see Groupoids, Definition 03BC and discussion following it). By construction $s_W, t_W : R_W \to W$ are finite étale. Consider the open subspace $X' = W/R_W \subset X$ (see Spaces, Lemma 02WU). By construction the inclusion map $X' \to X$ induces a bijection on points of codimension 0. This reduces us to Lemma 0BAS. □